

# Project Proposal

## Overview

This project focuses on preparing the CTU-IoT Malware dataset for machine learning. The dataset consists of labeled network traffic representing both benign and malicious activity, making it suitable for cybersecurity-focused analysis.

## Objectives

The objective is to transform raw network traffic data into a structured and analyzable format. This includes data cleaning, feature engineering, encoding categorical variables, scaling numerical features, and performing exploratory data analysis (EDA).

## Methodology

The dataset is split into training (70%) and testing (30%) sets. All EDA is performed exclusively on the training data to avoid data leakage. The testing dataset remains untouched for future evaluation.

## Expected Outcome

The result is a reproducible preprocessing pipeline that prepares real-world network data for machine learning models and highlights patterns between malicious and benign traffic.